Class Imbalance & SMOTE Strategy

- **Before SMOTE:** 517,897 legit (98.68%) vs. 6,949 phishing (1.32%)
- **After SMOTE (training only):** 414,317 legit vs. 414,317 phishing (synthetic + original)
- **Result:** Models now learn to detect phishing instead of predicting everything as "legit"

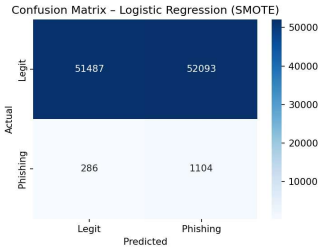
Model Comparison Results

Algorithm	Accuracy	F1-Score	Precision	Recall	ROC-AUC
■ Decision Tree	0.7284	0.0688	0.0361	0.7583	0.8128
Logistic Regression	0.5010	0.0404	0.0208	0.7942	0.6857
OLS Regression	0.5439	0.0319	0.0164	0.5683	N/A

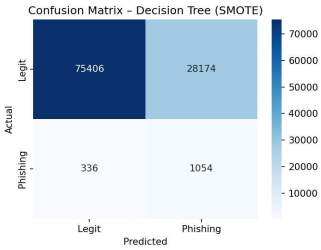
Winner: Decision Tree – Highest F1-Score (0.0688), Recall (75.8%), ROC-AUC (0.8128). Registered in MLflow Model Registry as **PhishingDetectionBestModel v2**

Model Performance – Confusion Matrices

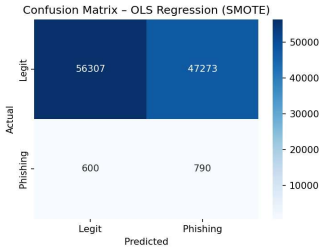
Logistic Regression



Decision Tree (Best)



OLS Regression



Decision Tree: 1,054 correct phishing detections (Recall 75.8%)
LogReg: 1,104 correct phishing detections (Recall 79.4%), but more false positives
OLS: Poor performance – unsuitable for this task

Confidence Intervals for OLS Coefficients

Feature	95% CI Lower	95% CI Upper
num_urgent_keywords	-0.05526	-0.05065
num_links	-0.00776	-0.00704
num_unique_domains	-0.00318	-0.00118
num_email_addresses	-0.00539	-0.00517

All intervals exclude zero → statistically significant effects. Negative coefficients indicate features that reduce phishing probability.

Key Findings

- **SMOTE was critical:** Without it, models achieved 98% accuracy but detected 0 phishing emails
- **Recall vs. Precision trade-off:** Decision Tree detects 75.8% of phishing but flags many false positives (3.6% precision)
- **Decision Tree outperforms:** Better at learning non-linear patterns than LogReg or OL